



ABSTRACTION

Vehicle Rental Management System (Week 1)



The Vehicle Rental Management System is designed to simplify and automate the process of managing vehicles in a rental business. The system abstracts the real-world rental operations by representing each vehicle as an object with essential details such as **Vehicle ID**, **Vehicle Name**, and **Rent Per Day**. Users interact with the system through a menu-driven console interface that provides high-level operations like **Add Vehicle**, **View Vehicles**, **Update Vehicle**, and **Delete Vehicle**, without needing to understand how the data is stored or processed internally.

The system maintains all vehicle records in an **ArrayList**, which stores multiple vehicle objects dynamically. Encapsulation is used to hide the internal data of each vehicle and provides access only through getter and setter methods, ensuring data security and consistency. By abstracting the complex data handling and focusing only on the necessary operations, the system makes it easy for users to manage vehicle information efficiently.

This abstraction reduces code complexity, improves organization, and enhances maintainability. It serves as a strong foundation for future enhancements such as customer management, vehicle booking, rental transactions, billing, and report generation. Overall, the Vehicle Rental Management System provides a simple, reliable, and user-friendly solution for managing rental vehicles effectively.